Year 2 Computing – Semester 3
COMP H2031: OOAD
Object Oriented Analysis & Design

Academic term	2024/25
Year of study	Year 2
Semester	SEMESTER ONE
Date of examination	
Time of examination	95 minutes + 5 minutes to upload = total 100 minutes

COMP H2031: Object Oriented Analysis & Design

Java Coding Lab Test 1

Internal Examiner(s)	**Orla McMahon, Krzysztof Szczurowski**

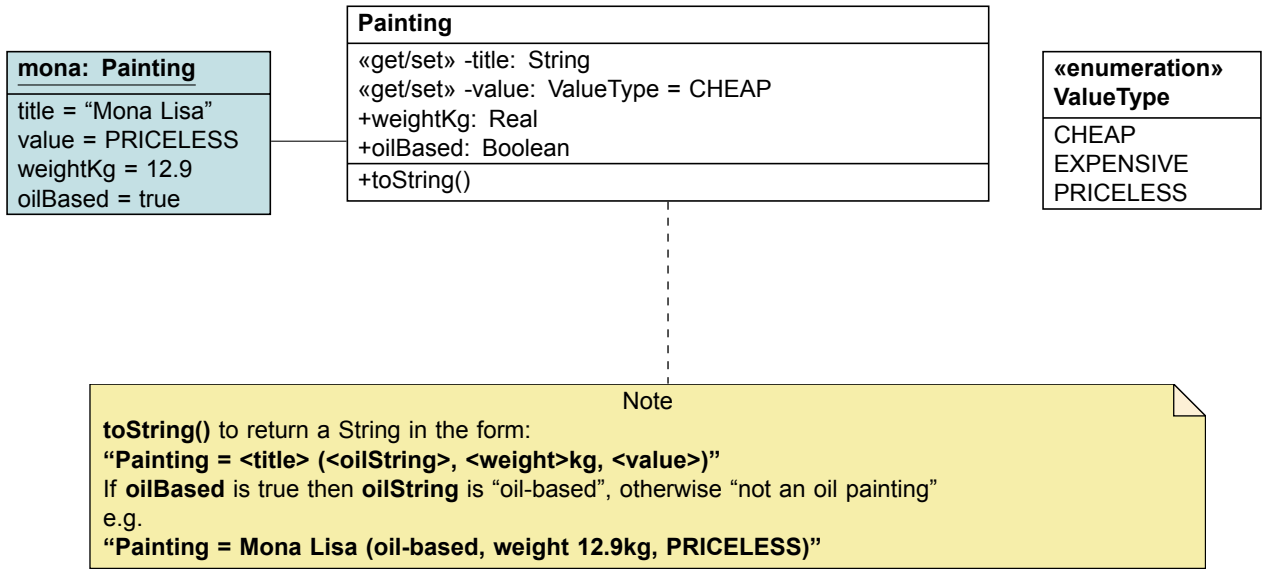
Instructions to candidates:

1.	**This is an individual test – all work should be your own.** **During the test you should NOT communicate with anyone apart from the lecturer (excepting medical / emergency circumstances).**
2.	**You must attend the test for your timetabled lab group.** **If you are not on the list for a group you will not be permitted to take the test.**
3.	**Bring physical photo ID: e.g. student card / driving licence / passport** **Place this beside you during the test.**
4.	**Use of headphones, phones, tablets, email etc. is NOT PERMITTED.**
5.	**This is an open-book test for paper-based materials.** **Use of GEN AI is not permitted**
6.	**The test is marked out of 100.** **Answer ALL questions.**

DO NOT TURN OVER THIS PAGE UNTIL YOU ARE TOLD TO DO

Question 1: Objects, enumeration & objects *Total: 30 marks*

Submit your answers to this question in folder: **question1/**



1a. Write Java files to implement the classes in the diagram above.

1b. Write a **main(...)** method in class **Main** to do the following:

- create and print out the details of the object(s) in the diagram.

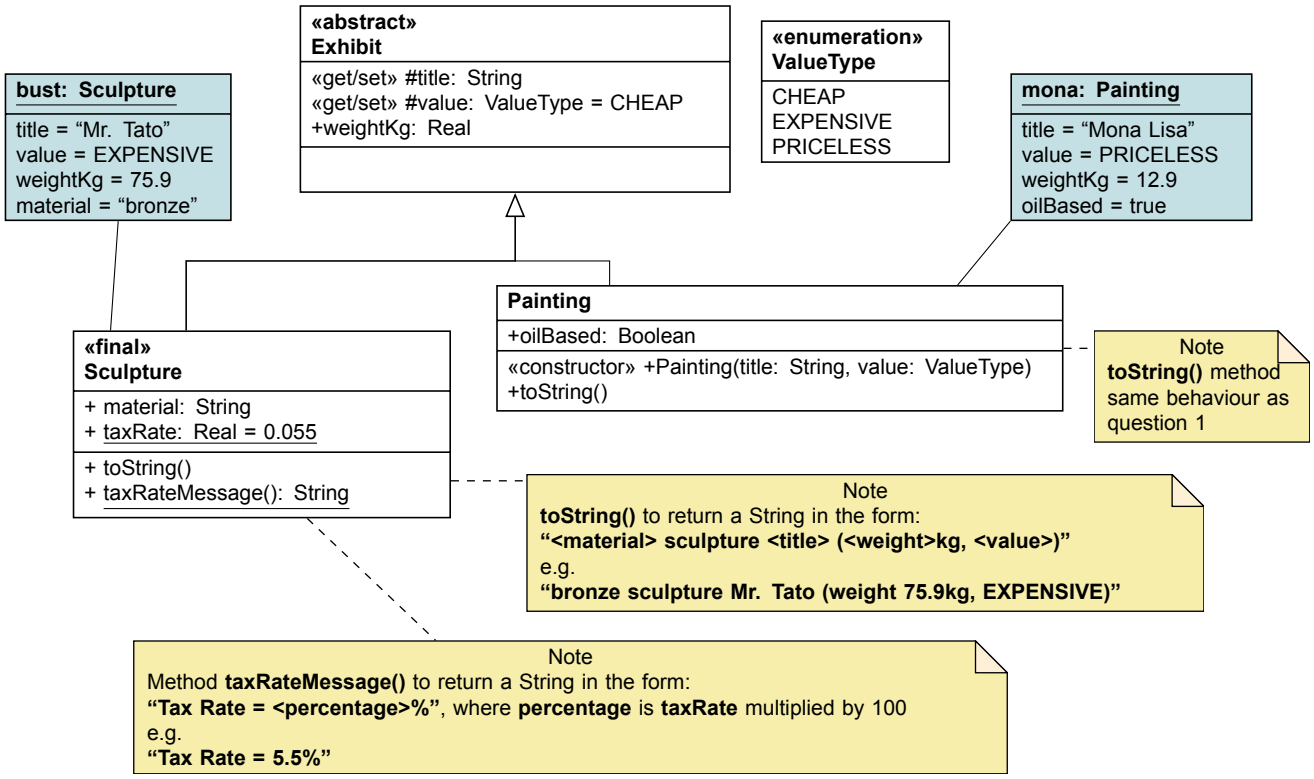
Your program output should look as follows:

```
$ java Main

Painting = Mona Lisa (oil-based, weight 12.9kg, PRICELESS)
```

Question 2: Inheritance, protected & static *Total: 70 marks*

Copy your question 1 folder, naming the copy: **question2/**. Adapt your code to answer this question.



2a. Write Java files to implement the classes in the diagram above.

2b. Write a **main(...)** method in class **Main** to do the following:

- Create the **Painting** object from the diagram, and print out its details
- Create the **Sculpture** object from the diagram, and print out its details
- Print out the value returned from **Sculpture** method **taxRateMessage()**
- Change the value of **taxRate** to **0.1**
- Print out the value returned from **Sculpture** method **taxRateMessage()**

Your program output should look as follows:

```
$ java Main

Painting = Mona Lisa (oil-based, weight 12.9kg, PRICELESS)

bronze sculpture Mr. Tato (weight 75.9kg, EXPENSIVE)

Tax Rate = 5.5%

Tax Rate = 10%
```